

Michael Mulugeta Demissie

Data Engineer | ML Infrastructure · ETL Pipelines · Cloud · Python

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SUMMARY

Data Engineer with 3+ years of hands-on experience building production ETL pipelines, deploying ML models, and shipping full stack data applications. Skilled across the full data lifecycle, from ingestion and transformation to model serving and API delivery.

EDUCATION

The George Washington University | M.S. Data Analytics

Expected May 2027

Delhi Technological University | B.Tech Software Engineering

May 2025

PUBLICATIONS

Cost-Effective Real-Time Targeted Advertising on Public Screens

Jan 2026

Springer Nature | 7th Intl. Conference on Smart Computing and Informatics (SCI 2025)

- Co-authored Springer paper on real-time ML pipeline integrating CV and weather data for dynamic targeted advertising (SCI 2025).

TECHNICAL SKILLS

Data Engineering: Python, SQL, Airflow, Spark, Kafka, dbt, ETL/ELT, Data Modeling

Data Analytics: Pandas, NumPy, Tableau, Power BI, Plotly, Dash, Statistical Analysis

Cloud & Infra: AWS (S3, Glue, Redshift, Lambda), Snowflake, BigQuery, Terraform, Docker

Databases: PostgreSQL, MySQL, Redis, Elasticsearch, MariaDB

ML & AI: PyTorch, TensorFlow, scikit-learn, HuggingFace, BERT, OpenAI API, MLflow, Groq, pgvector, RAG

Full Stack: FastAPI, Flask, Node.js, Express, React, JavaScript, Bash, Git, CI/CD, JWT Auth

EXPERIENCE

Data Engineer | Hasab AI

May 2025 – Apr 2026

Remote

- Architected Python ETL pipelines processing multilingual audio datasets across 10+ languages · segmentation, noise filtering, and feature extraction.
- Built FastAPI services exposing structured NLP data to downstream speech and voice AI applications.

Data Engineer | ParallelDots

Aug 2024 – Jun 2025

Delhi, India

- Engineered Python ETL pipelines ingesting retail shelf image metadata into MySQL, powering the ShelfWatch AI CV platform used by Fortune 500 FMCG clients globally.
- Built Pandas/NumPy preprocessing pipelines for CV model training and developed Flask REST APIs exposing processed data to production dashboards.

TECHNICAL PROJECTS

FinResolve: ETL Pipeline & ML Routing | Live | GitHub

Jan 2026 – May 2026

Python · Airflow · PostgreSQL · AWS S3 · Docker · FastAPI · DistilBERT · Render

- Built end-to-end ETL pipeline processing 1M+ CFPB records; engineered 812K training samples across 9 departments for ML routing.

DC Metro Real-Time Arrivals Pipeline | GitHub

May 2026

Node.js · PostgreSQL · Airflow · Docker · Express · Railway · WMATA API

- Built real-time pipeline polling WMATA API every 60s, sustaining 580K+ rows/day into PostgreSQL with Airflow 24hr retention.

DocuQuery: Production RAG & AI Document Search Pipeline | Live | GitHub

May 2026 – Jun 2026

Python · FastAPI · PostgreSQL · pgvector · HuggingFace · Groq LLM · Airflow · Docker · Railway

- Built end-to-end RAG pipeline: document ingestion, chunking, HuggingFace embeddings, pgvector storage with HNSW index, and sub-second semantic search.
- Deployed via Docker with FastAPI streaming, Groq LLM, Airflow DAG ingesting 85+ daily federal job postings, and session-isolated document storage on Railway.